

M9 — Student join flow (room QR → roster → encounter → station)

Phase: 5 — Roster **Status:** DONE (2026-05-26) **Blocked by:** M6, M8 **Blocks:** M10, M20 **Estimated effort:** 2 days · **Actual:** 0.5 day

1. Purpose

The student-side journey: scan a room QR → land on `/portal/students/join?code=ROOM_CODE` → pick a name (from the operator-loaded roster) or type a free-form display name → see the room's encounter cards → tap one → join an encounter's chat station as the next student. The encounter's patient persona is the conversational partner (room-mode encounters have exactly one).

This is the missing wire between the M8 student-roster persistence layer and the existing v6 chat-station UI. After M9, the operator flow stays in the v7 dashboard, students join through M9's public page, and both meet at the existing v6 `/station/{join_code}/{station_id}` chat UI.

2. Structure

New files:

- `portal/templates/student_join.html` — Public landing page, extends nothing (no operator nav). Three states: error, step-1-name, step-2-encounter.
- `portal/static/student_join.css` — Mobile-first; large tap targets; roster grid + encounter grid responsive.
- `portal/static/student_join.js` — Roster pick / free-form name capture; encounter card POSTs `/portal/students/register` with the form fields; redirects on success.

Files touched:

- `portal/server.py` — adds two routes under "V7 — Student join flow (M9)": - GET `/portal/students/join?code=ROOM_CODE` (public) - POST `/portal/students/register` (public) And a helper `_room_by_code(room_code)` that resolves a room by its operator-displayed code (case-insensitive).

3. Uses

- The M5 charge-nurse dashboard shows the room code; the operator reads it aloud, or displays the QR (M21 LAN test step). Students scan/type the code.
- Students arrive at `/portal/students/join?code=<code>` → see encounter cards + (optionally) the pre-loaded roster.
- They tap an encounter → JS posts `/portal/students/register`:
- `room_code`, `encounter_id`, `display_name` (required for new), `existing_student_id` (optional — reattach).
- The handler:

1. `room = _room_by_code(...)` — 404 if unknown
 2. `room.add_student(display_name)` OR `reattach_via_existing_student_id`
 3. `room.assign_student(student_id, encounter_id)` → M8 writes through to DB.
 4. `enc.add_station(station_id)` with `enc.patient_persona_id`.
 5. Returns `{redirect_url: "/station/{enc.join_code}/{station_id}"}`.

- The JS redirects → the existing v6 chat-station UI loads with

the patient persona ready to chat.

4. Functions (exported API surface)

HTTP routes

Method	Path	Auth	Returns
GET	<code>/portal/students/join?code=ROOM_CODE</code>	public	HTML template (error / step-1 / step-2 states)
POST	<code>/portal/students/register</code>	public (room_code is the access token)	<code>{ok, student_id, display_name, encounter_id, station_id, redirect_url}</code>

POST body

Field	Required	Notes
<code>room_code</code>	yes	Case-insensitive; uppercased server-side.
<code>encounter_id</code>	yes	Must be an id in <code>room.encounters</code> .
<code>display_name</code>	yes (for new students)	Free-form.
<code>existing_student_id</code>	no	Reattach to an existing Student row (no duplicate).

Errors

Status	When
400	<code>display_name</code> blank AND no <code>existing_student_id</code> .
404	Unknown <code>room_code</code> , unknown <code>encounter_id</code> , unknown <code>existing_student_id</code> .
409	Encounter has no patient persona configured.

Internal helpers

Symbol	Purpose
<code>_room_by_code(room_code)</code>	Resolve a room by operator-displayed code. Case-insensitive match against the active room. Returns None on miss.

5. Limitations

- **One active room at a time.** Matches the single-instructor model.

If the operator ends room A and starts room B with a colliding 6-char code, students scanning A's stale code will see B's content. The 6-char alphabet has ~387M codes — collision is improbable but possible. M21 LAN test

should verify.

- **No persona pick.** Room-mode encounters have one

`patient_persona_id`; the student doesn't choose. In single- patient mode (rooms-of-1 created by the v6 wizard branch), the encounter may carry multiple `selected_personas` — the M9 handler falls back to the first one when `patient_persona_id` is None. For multi-persona conversations the v6 `/join?code=` flow is still the right path.

- **Roster reattach by existing_student_id only.** No "match by

`display_name`" — that would silently re-bind a student to a stale row. The roster-card UI passes the `student_id` explicitly when the student taps a pre-loaded name.

- **No heartbeat yet.** M8's `touch_student` helper exists but no

route calls it. M16's WebSocket transport will add a heartbeat channel.

- **No instructor override.** A student picks themselves; the

operator can't force-rebind without going through the M4 `/api/encounter/{id}/assign_students` route (which doesn't yet write through to the DB — see M8 known issues §8).

- **Cookies not used.** Each tab is a fresh student. A future

v7.1 may set a student session cookie so a refresh or short- duration disconnect doesn't require re-entering the name. For now, the chat station's existing `station_id` in the URL persists the identity.

6. Test status

Automated (tests/v7/test_student_*.py)

Test file	Cases	Status	Last run
<code>test_student_register_then_pick_encounter.py</code>	5 — happy path register+assign +station+redirect, reattach to existing roster entry, GET page renders room + roster, blank name 400s, unknown encounter 404s.	PASS	2026-05-26
<code>test_student_join_url_handles_unknown_room_code.py</code>	6 — no code → error form, unknown code → error, stale code after room swap, lowercase code, POST 404s on unknown code, POST 404s with no active room.	PASS	2026-05-26

11/11 PASS. **Full v7 suite: 54/54 passing** (up from 43). **Full v6 regression on v7: 165 passed**, same 6 env-flaky pre-existing failures, **0 v7 regressions**.

Manual (browser preview — 2026-05-26)

Flow	Result
GET <code>/portal/students/join?code=<code></code> after starting a 2-bed room	PASS — room code visible, no operator chrome; step 1 (name) shown; step 2 (encounters) hidden until name is entered.
Type name + blur	PASS — step 2 reveals with "Welcome, Alice Pham. Pick a bed to start." status.

Flow	Result
Tap Bed 1 card	PASS — POST register, redirect to /station/6NDBQF/<sid> with v6 station UI loaded; persona = encounter's patient_persona_id.
Operator dashboard polls after join	PASS — Bed 1 chat_stations: 0→1, students: 0→1; Bed 2 untouched (encounter scoping holds).

7. Change list

Date	Author	Change	Files
2026-05-26	claude-corde	Two new public routes (GET /portal/students/join, POST /portal/students/register) + _room_by_code helper. New public template + CSS + JS for the student-side join page. 11 acceptance tests across 2 files. Live browser verification of the operator→student handshake.	portal/server.py, portal/templates/student_join.html, portal/static/student_join.{css,js}, tests/v7/test_student_register_then_pick_encounter.py, tests/v7/test_student_join_url_handles_unknown_room_code.py

8. Open questions / known issues

- The M4 dashboard's /api/encounter/{id}/assign_students route

still mutates state directly (not through `room.assign_student`), so DB write-through doesn't fire on that path. M9 doesn't fix this — its own register flow uses the correct helper. Migrating M4's route is a small cleanup; the contract for M9 is met.

- No "back" button on the student page after picking an encounter.

If the student taps the wrong bed, they'd have to ask the instructor to reassign. Adding a back affordance is a v7.1 UX polish.

- The page doesn't surface "this room is frozen" — a frozen room

still lets students join (they'll just see a paused chat station). Whether to block joins during freeze is a policy question; M5 dashboard treats freeze as "everyone hold," so blocking joins is the natural extension. Defer until M21 LAN feedback clarifies the operator intent.

- QR code generation for the room code itself is M5 follow-up

(the dashboard currently just displays the code in text). M21 LAN test exercises a real QR scan; if a tap-friendly QR is missing, surface it on the dashboard then.

- The 6-char `room_code` alphabet excludes visually similar chars

(0, O, 1, l, L), giving ~387M codes — collisions are improbable but possible. M19 capacity hardening should add a uniqueness check across recent ended rooms if collisions ever materialize.